

Deepak Mallampati

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Kent, OH - F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer with 2.5+ years delivering agentic LLM systems, multi-provider API orchestration, and production RAG pipelines in regulated healthcare. Shipped full-stack browser extensions, multimodal API platforms, and multi-agent reasoning systems end-to-end. Translates complex AI requirements into working integrations quickly - the core capability of a Forward Deployed Engineer. Comfortable working across TypeScript, Python, FastAPI, Docker, and LLM provider APIs. US-based on F-1 OPT, seeking sponsorship.

CORE COMPETENCIES

Agentic LLM Workflows • Retrieval-Augmented Generation (RAG) • API Orchestration • TypeScript / React • Python / FastAPI / Flask • Multi-Provider AI Integration • Structured Outputs & Prompt Engineering • Docker / CI/CD • LLMOps • Healthcare AI (HIPAA-conscious) • Client-Facing Delivery • Fast Prototyping

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS

- Built **production RAG systems** for clinical knowledge retrieval with recursive semantic chunking and transformer embeddings; improved contextual retrieval precision **~35%**, reduced irrelevant retrieval **>30%**, retrieval-grounded response alignment exceeded **90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning and tool discipline; improved agent response stability **~25%** and reduced hallucinated outputs **>30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall **15-20%** on high-risk categories at **>90%** precision.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services with Docker, structured logging, and load simulation; reduced post-deployment defects **~30%**.
- Maintained HIPAA-conscious data governance with de-identification, data lineage, and evaluation audit trails.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP

- Optimized T-SQL stored procedures for Order-to-Cash platform; reduced query latency **~25%** and execution time **~20%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and SQL deployments; reduced deployment errors **>30%**, improved release cycle **~35-40%**.
- Designed Grafana dashboards tracking long-running queries, deadlocks, and I/O contention; cut incident recurrence **~25%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics

- Built transformer-based video summarization system across 5,000+ sessions; **60-70% reduction** in manual review time, **~85% alignment** with human-curated highlights.
- Implemented document Q&A with RAG; reduced hallucinated outputs **~30%**, raised contextual accuracy **>85%**.

Kent State University

Aug 2023 - May 2025

Student Manager - Operations, Team Leadership

- Led coordination for **150+ students**; reduced coordination delays **~30%** and formalized onboarding cutting ramp-up **~20%**.

FEATURED PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Translation shipped, Chrome Web Store

Full-stack Manifest V3 extension integrating four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider webhook-style routing with per-domain selector configs for 29 sites, seven-day translation cache, and viewport-slice screenshot pipeline with coordinate remapping.

TypeScript, React, Manifest V3, multi-provider API orchestration, WebP/JPEG routing

Dream Decoder - Multimodal AI Creative Platform

FastAPI + React/TypeScript/Vite/Tailwind app coordinating multiple AI APIs (Perplexity Sonar, Replicate image models). Intermediate structured prompt transformation layers improved contextual alignment ~**30%** and first-pass image success rate ~**25-30%** over naive direct-prompt approaches.

FastAPI, React, TypeScript, multi-LLM orchestration, structured outputs, REST APIs

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate detection; explainable risk scoring with audit-ready traces.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG, agentic orchestration

Agentic Pixel Character Synthesis & Animation Engine

Phased generative AI system for identity-consistent pixel character synthesis. Stable Diffusion fine-tuning + LoRA, ControlNet + OpenPose/MediaPipe for pose conditioning, and an **LLM-based agent orchestrator** decomposing high-level prompts into generation subtasks.

PyTorch, Hugging Face Diffusers, ControlNet, FastAPI, Docker, LLM orchestration

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages: Python, TypeScript, JavaScript, FastAPI, Flask, React, SQL

LLM & GenAI: LLM APIs (Claude, GPT-4o, Gemini), RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines

AI Libraries: LangChain, LlamaIndex, Hugging Face Transformers, scikit-learn, PyTorch

Integration & APIs: RESTful APIs, WebSockets, webhooks, multi-provider routing, OAuth, Manifest V3

Infrastructure: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, cloud-ready deployment